

PATIENT HISTORY:

CHIEF COMPLAINT/ PRE-OP/ POST-OP DIAGNOSIS: Papillary thyroid carcinoma.
 PROCEDURE: Total thyroidectomy.
 SPECIFIC CLINICAL QUESTION: Not provided.
 OUTSIDE TISSUE DIAGNOSIS: Not provided.
 PRIOR MALIGNANCY: Not provided.
 CHEMORADIATION THERAPY: Not provided.
 ORGAN TRANSPLANT: Not provided.
 IMMUNOSUPPRESSION: Not provided.
 OTHER DISEASES: Not provided.

100-0-3
 carcinoma, papillary, tall cell variant 8344/3
 Site: thyroid, NOS C73.9 w 11/27/11

ADDENDA:**Addendum****Molecular Anatomic Pathology Testing:****Block 2C:**

- A. **BRAF V600E mutation IDENTIFIED.**
- B. Mutations in **NRAS61, HRAS61, KRAS12/13** NOT identified.
- C. FISH test results are negative for **RET/PTC** rearrangement.

Criteria	Yes	No
Diagnosis Discrepancy		☒
Primary Tumor Site Discrepancy		☒
HIPAA Discrepancy		☒
Prior Malignancy History		☒
Dual/Synchronous Primary Sites		☒
Case is (circle):	QUALIFIED	DISQUALIFIED
Reviewer Initials	WJ	11/27/11

Note:

DNA was extracted in the amount sufficient for testing.

Mutations in either **BRAF** or **RAS** genes or **RET/PTC** rearrangements are found in more than 70% of papillary thyroid carcinomas (1). **BRAF** mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between **BRAF V600E** mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the **RAS** genes or **PAX8/PPARg** rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, **BRAF V600E** mutation and **RET/PTC** and **PAX8/PPARg** rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). **RAS** mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

1. Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006, 30:216-222.
2. Xing M, et al. **BRAF** mutation predicts a poorer clinical prognosis for papillary thyroid cancer. *J Clin Endocrinol Metab* 2005, 90:6373-6379.
3. Elisei R, et al. **BRAF V600E** mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008, 93:3943-3949.
4. Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential **BRAF (V600E)** mutational status in high aggressive papillary thyroid microcarcinoma. *Ann Surg Oncol* 2009 Feb;18(2):240-5.
5. Nikiforova MN, et al. **RAS** point mutations and **PAX8-PPARg** rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003, 88: 2318-26.
6. Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV., ed. *Endocrine Pathology: Differential Diagnosis and Molecular Advances*. Humana Press, Totowa, 2004, 191-209.

Sample Preparation and Procedure

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the platform to amplify **BRAF**, **NRAS** codon 61, **HRAS** codon 61, and **KRAS** codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. If required, the mutation type was confirmed by Sanger sequencing of the PCR product on DNA from samples positive for each of these mutations was used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10% of alleles with mutation present in the background of normal DNA and RNA.

FINAL DIAGNOSIS:**PART 1: PYRAMIDAL LOBE, EXCISION -**

BENIGN THYROID TISSUE WITH NODULAR HYPERPLASIA.

PART 2: THYROID, TOTAL THYROIDECTOMY (69.7 GRAMS) -

- A. **PAPILLARY THYROID CARCINOMA, 6.0 CM, TALL CELL VARIANT, LOCATED IN LEFT LOBE.**
- B. **EXTRACAPSULAR EXTENSION IS PRESENT.**
- C. **SURGICAL RESECTION MARGINS ARE NEGATIVE, ALTHOUGH TUMOR IS SEEN 0.1 CM FROM MARGINS.**
- D. **ONE LYMPH NODE WITH METASTATIC CARCINOMA (1/1).**
- E. **BACKGROUND THYROID PARENCHYMA WITH NODULAR HYPERPLASIA.**
- F. **AJCC TUMOR STAGE: pT3 N1a MX.**

PART 3: LYMPH NODES, CENTRAL COMPARTMENT, DISSECTION -

THREE OUT OF FOUR LYMPH NODES WITH METASTATIC CARCINOMA (3/4).

COMMENT:

Molecular studies are ordered and results will be reported in an addendum.

UUID: F4DC EE4E-CB1C-48CB-8BE2-90A9925DBA70
 TCGA-B1-A0Z5-01A-PR

Redacted



CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE: Total Thyroidectomy
TUMOR SITE: Left Lobe ✓
TUMOR FOCALITY: Unifocal
TUMOR SIZE (largest nodule): Greatest Dimension: 6.0 cm
HISTOLOGIC TYPE: Papillary carcinoma, tall cell variant ✓
PATHOLOGIC STAGING (pTNM): pT3
pN1a
Number of regional lymph nodes examined: 5
Number of regional lymph nodes involved: 4
pMX
MARGINS: Margins uninvolved by carcinoma
Distance of invasive carcinoma to closest margin: 1 mm
VENOUS/LYMPHATIC (LARGE/SMALL VESSEL) INVASION (V/L): Absent
ADDITIONAL PATHOLOGIC FINDINGS: Other: Nodular hyperplasia

SPECIAL PROCEDURES:

In Situ Procedure

Interpretation

RESULTS:

RET/PTC: The targeted area of the tissue showed 0(0%) of the cells with the rearrangement pattern and 60(100%) of the cells with the normal pattern. The targeted area is considered NOT rearranged for the *RET/PTC* region.

The *RET/PTC1* Probe did not contain the rearrangement pattern.

Results

RET/PTC Methodology Description:

Probe: *RET/PTC* Probe (10q11.2) and *RET/PTC1* Dual-Color Fusion Probe 10q11.2-10q21

Probe Description: The *RET* probe (labeled) is a 207 kb DNA probe located at band 10q11.2.

The H4 and 369 probes are approximately 100kb each (labeled in - -), and are located at band 10q21.

RET/PTC rearrangement is suggested to be present when three *RET* signals are identified in greater than 8% of the analyzed cells.

RET/PTC1 rearrangement detection is as follows: Normal cells without the *RET/PTC1* rearrangement show two pair of green and orange signals separated. Cells with the *RET/PTC1* rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

RET/PTC FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells using the *PTC* (369/H4) SpectrumOrange and the *RET* SpectrumGreen probes.